



YACHAY TECH UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

LABORATORY REPORT

Title of the Report

Technical Report

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COURSE Cryptography

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1 Introduction

Provide a concise introduction to the problem, project, or topic addressed in this report. Explain the context and why the work is relevant.

2 Objectives

2.1 General Objective

State the main objective of the project.

2.2 Specific Objectives

- Specific objective one.
- Specific objective two.
- Specific objective three.

3 Background

Present the concepts, technologies, algorithms, or theoretical foundations required to understand the project. Cite every external source, as in [1].

3.1 Concept One

Explain the first relevant concept.

3.2 Concept Two

Explain the second relevant concept.

$$a \equiv b \pmod{n} \tag{1}$$

4 Methodology

Describe the methodology, tools, environment, and procedures used.

4.1 Tools and Technologies

Table 1. Tools and technologies used in the project.

Tool	Purpose
Linux	Development and execution environment.
Python	Implementation of project components.
Docker	Containerization.
Git	Version control.

4.2 Procedure

Describe the steps followed to complete the project.

5 Development

Describe the implementation of the proposed solution.

5.1 System Architecture

Explain the architecture and the relationship between its components.

5.2 Implementation

Describe the relevant implementation details.

5.2.1 Example Code

The following listing illustrates a code fragment used in the project.

Listing 1. Example Python implementation.

```

1 def main():
2     print("Hello, World!")
3
4 if __name__ == "__main__":
5     main()

```

As shown in listing 1, the implementation follows a simple entry-point structure.

6 Results

Present the results obtained during the implementation or experimentation.

6.1 Experimental Results

Explain the observations and measurements obtained, as summarized in table 2.

Table 2. Example results.

Experiment	Result	Status
Test 1	95%	Successful
Test 2	91%	Successful
Test 3	87%	Successful

6.2 Discussion

Interpret the results and explain their relevance.

7 Conclusions

Summarize the principal findings of the work. Relate the conclusions directly to the objectives defined at the beginning of the report.

8 Future Work

Describe improvements, extensions, or research directions that could be implemented in future versions of the project.

References

- [1] J. Katz and Y. Lindell, *Introduction to Modern Cryptography*, 3rd ed. Boca Raton, FL, USA: CRC Press, 2020.

A Additional Material

Place supplementary information here, such as extended code, additional tables, configuration files, screenshots, or detailed calculations.